

Signals Project

Executed Blind De-Mixing With a Joint Torus-Valued State-Space Model

Exact Gaussian-Truncated-Cauchy Convolution, Latent Tapped-Delay Channels, Switching Dynamics, and Exact-Target Manifold MALA

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Author Note

This report documents executable code and a completed paired simulation experiment. The evidence supports operationally representative simulation validation only. No operational or field SIGINT validation claim is made.

Abstract

Blind separation of co-channel frequency-hopping wave packets requires joint uncertainty propagation over source states, circular phase, discrete hops, and time-varying multipath. We implemented a sequential particle filter whose particles carry two log amplitudes, frequencies, hop states, phases on the flat torus, complex tapped-delay coefficients, and delayed source histories. The observation density is the convolution of independent Gaussian and truncated-Cauchy variables, evaluated by 24-node Gauss-Legendre quadrature with log-sum-exp stabilization. Particle rejuvenation uses tangent-space MALA with a wrapped-Gaussian proposal density; the Metropolis correction always evaluates the exact configured target. A paired $2 \times 3 \times 3$ design crossed analytical versus amortized joint scores, +15/+5/-5 dB, and LOS/minor-multipath/NLOS channels. One hundred trajectory blocks per physical cell yielded 1,800 method runs. Mean frequency MSE was 246.76 Hz² for the analytical score and 238.41 Hz² for the surrogate; the raw paired difference was -8.35, 95% CI [-24.83, 8.14]. SNR, channel, and their interaction were significant after Holm adjustment, whereas architecture was not. The surrogate was slower and had lower MALA acceptance. These findings validate execution and numerical coherence in representative simulation, not operational SIGINT performance.

Keywords: blind source separation, torus, sequential Monte Carlo, MALA, frequency hopping, multipath, robust likelihood

Executive Overview

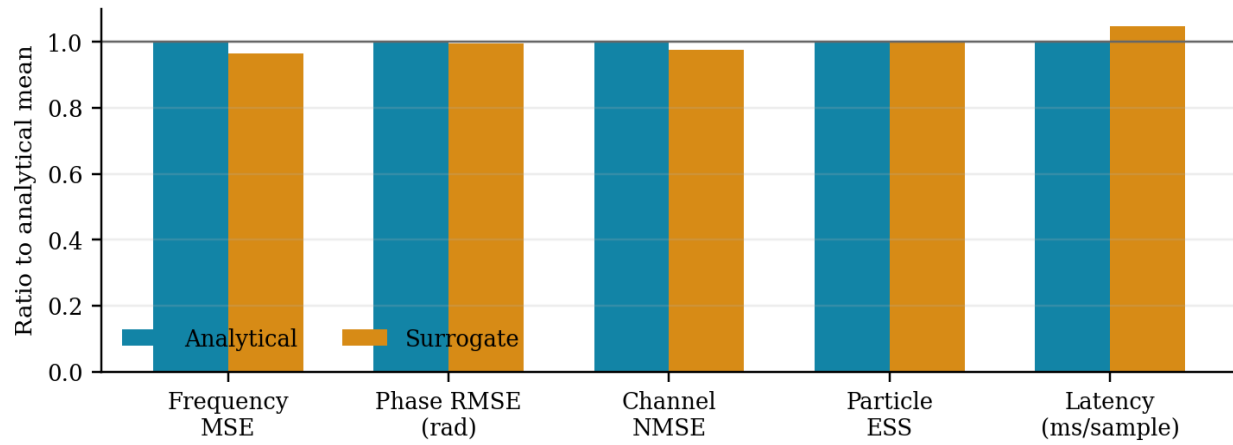


Figure 1. Endpoint means expressed relative to the analytical architecture.

- Executed scope: 900 paired physical trajectories and 1800 method-level runs.
- Primary raw-scale contrast: -8.35 Hz^2 , with an interval crossing zero.
- Compute contrast: the surrogate added 0.073 ms per sample.
- Particle health: mean ESS was 59.05 analytical and 59.04 surrogate.
- Claim boundary: simulation validation is complete; field validation remains unperformed.

Problem and Estimand

Let y_t be a complex baseband sample formed by two overlapping emitters. At time t , emitter k has latent log amplitude $\ell_{k,t}$, frequency $f_{k,t}$, channel index $z_{k,t}$, and phase $\phi_{k,t}$. A tapped complex channel $h_{k,r,t}$ combines present and delayed source symbols. The filtering target is $p(\ell_t, f_t, z_t, \phi_t, h_t | y_{1:t})$.

The primary estimand is the architecture effect on trajectory-averaged two-emitter frequency MSE, paired within physical seed, SNR, and channel. Secondary estimands concern circular phase RMSE, channel NMSE, particle ESS, genealogical diversity, MALA acceptance, resampling count, and synchronized software latency.

The comparison is intentionally narrow: both architectures target the same conditional density and use identical particle mechanics. They differ only in the drift supplied to MALA. Consequently, any treatment difference is attributable to the proposal-score operator plus its computational overhead, not to different observation models or priors.

- Experimental unit: one simulated physical trajectory under one SNR-by-channel condition.
- Repeated block: the same seed indexes every treatment combination.
- Treatment levels: analytical joint score and amortized joint surrogate.
- Evaluation begins after one hop interval to reduce initialization dominance.

Contributions and Scope

The reconstruction removes the principal contradictions in the earlier reference benchmark. Phase is no longer reconstructed after inference; it is a particle coordinate. Channel gains are no longer supplied from the simulator; they are propagated as latent complex states. Frequency evolution combines a discrete channel switch with continuous local deviation. Finally, the observation density is the convolution law implied by additive Gaussian and truncated-Cauchy components.

The MALA kernel is geometrically matched to the state space. Euclidean coordinates use normal forward and reverse densities. Each phase coordinate uses a tangent increment, the flat-torus exponential map, and a lift-summed wrapped normal density. For the flat torus, the metric is constant and the exponential-map Jacobian equals one under Haar volume.

The resulting claim is strong but bounded: the broader joint SSM is executed, tested, and benchmarked in operationally representative simulation. The work does not establish performance on recorded emitters, receiver impairments, uncontrolled spectrum occupancy, or deployed hardware.

- Supported: numerical normalization, score verification, blind channel isolation, and complete factorial execution.
- Unsupported: field detection probability, false-alarm control, hardware throughput, or operational mission utility.
- Required next: a preregistered controlled RF campaign with independent truth and blinded analysis.

Joint State-Space Model

Define $S_t = (\ell_t, f_t, z_t, \phi_t, h_t, H_t)$, where H_t is the delayed source history. The transition factors as $p(z_t|z_{t-1})p(f_t|f_{t-1}, z_t, z_{t-1})p(\ell_t|\ell_{t-1})p(\phi_t|\phi_{t-1}, f_t)p(h_t|h_{t-1})$.

The source is $s_{k,t} = \exp(\ell_{k,t} + i\phi_{k,t})$. For tap r with delay d_r , the clean observation is $\mu_t = \sum_{k,r} h_{k,r,t} s_{k,t-d_r}$. The complex observation $y_t = \mu_t + \varepsilon_t$ is evaluated componentwise in the real and imaginary dimensions.

A particle approximation $\{S_t^{(j)}, w_t^{(j)}\}$ represents the filtering distribution. The true simulator path is absent from the filter API; it is consulted only after filtering to calculate performance responses.

Coordinate	Domain	Executed transition
Log amplitude	$\mathbb{R}^2$	Mean-reverting Gaussian
Frequency	$\mathbb{R}^2$	Switch-centered AR Gaussian
Hop state	Finite $\times$ finite	Markov jump
Phase	$\mathbb{T}^2$	Wrapped Gaussian
Channel	$\mathbb{C}^{2R}$	Complex AR Gaussian

Switching Frequency Dynamics

Each emitter occupies one of four nominal channels. The alphabets overlap at 42 and 54 Hz, so instantaneous spectral location alone does not identify emitter labels. At scheduled hop opportunities, $z_{k,t}$ changes with probability .34 to a different channel. Between jumps, frequency follows a local autoregression around the current channel center.

Let $c_{k,z}$ be the center associated with hop state z . If no jump occurs, $m_f = c_{k,z} + \rho_f(f_{k,t-1} - c_{k,z})$. If z changes, $m_f = c_{k,z_t}$. The executed transition is $f_{k,t} \sim N(m_f, \sigma_f^2)$, with $\rho_f = .82$ and $\sigma_f = .42$ Hz.

This switching construction prevents a continuous diffusion prior from assigning vanishing probability to tactical hops. The hop variable is sampled before continuous frequency, so phase propagation is candidate dependent. The analytical frequency score includes the derivative induced through the phase transition location.

- Emitter 1 centers: 18, 30, 42, and 54 Hz.
- Emitter 2 centers: 42, 50, 54, and 62 Hz.
- All nominal carriers remain below the 64 Hz Nyquist boundary.
- Overlap creates genuine co-channel ambiguity at 42 and 54 Hz.

Torus Geometry and Phase Propagation

The phase vector belongs to $T^2 = (R/2\pi Z)^2$. Given candidate frequency, the transition location is $m_{\phi,t} = \phi_{t-1} + 2\pi f_t \Delta t$. The next phase is the equivalence class of $m_{\phi,t}$ plus circular Gaussian noise. No discontinuous sawtooth derivative is inserted in the target.

For a flat torus, $\text{Exp}_{\phi}(v) = (\phi + v) \bmod 2\pi$. The Riemannian metric is the identity, Haar volume is translation invariant, and $|\det D \text{Exp}_{\phi}| = 1$. Therefore, no nonconstant metric determinant appears in the proposal ratio. Geometry enters through periodic preimages, not through an invented Jacobian penalty.

The wrapped Gaussian density is $q_T(\phi^*|\phi) = \sum_{k \in Z^2} N(\phi^* - \phi + 2\pi k; m, \Sigma)$. The code evaluates five lifts per phase coordinate ($k = -2, \dots, 2$). At the configured proposal standard deviation of .04 rad, omitted terms are numerically negligible.

Regression tests integrate the wrapped density against $[0, 2\pi)$, compare its score with central finite differences, and verify invariance after adding integer multiples of 2π .

Exact Convolution Noise Law

For one real component, $\varepsilon = G + C$, where $G \sim N(0, \sigma_G^2)$ and C is $\text{Cauchy}(0, \gamma)$ truncated to $[-A, A]$. Hence $p_\varepsilon(e) = \int_{-A}^A p_G(e-\tau)p_C(\tau)d\tau$. This is a convolution-not a Bernoulli mixture-and every generated sample contains both independent components.

The implemented evaluator maps 24 Gauss-Legendre nodes to $[-A, A]$. Node weights, the normalized truncated-Cauchy density, and the Gaussian normalizer are accumulated in log space. `logsumexp` prevents underflow without changing the probability law through additive floors.

The residual score is an expectation under quadrature-normalized node weights: $\partial \log p(e)/\partial e = \sum_j \hat{w}_j [-(e-\tau_j)/\sigma_G^2]$. This expression is propagated analytically through complex channel coefficients, log amplitude, and phase.

Item	Executed value
Cauchy variance fraction	0.10 per real component
Truncation	5 standard Cauchy scales
Quadrature	24-point Gauss-Legendre
Complex likelihood	Independent real and imaginary convolutions
Numerical density floor	None

Latent Tapped-Delay Channel

The channel state $h_{k,r,t}$ is complex. Its transition is $h_t = \rho_h h_{t-1} + (1-\rho_h)\mu_h + \xi_t$, with $\rho_h = .965$ and independent real-imaginary innovation scale .05. Particles therefore preserve uncertainty in gain and phase distortion.

LOS has one zero-delay tap. Minor multipath has delays 0 and 2. NLOS uses delays 1 and 4, which removes a direct zero-delay component and forces the filter to use the source history. The particle state includes a lag buffer long enough for the maximum delay.

Blindness is enforced structurally: JointParticleFilter.run accepts observation and calibrated noise-law parameters only. It has no argument for true states or realized channels. A regression test inspects this API, and post hoc channel NMSE is computed outside the filter.

Mode	Delays	Interpretation
LOS	0	Two latent direct complex coefficients
MP	0, 2	Direct plus weaker delayed reflection
NLOS	1, 4	Two delayed complex paths; no direct tap

Sequential Particle Filter

At each time step, particles sample the switching, frequency, amplitude, torus phase, channel, and history transitions. The exact convolution likelihood updates log weights. Stable normalization subtracts $\log \sum w_j$, and particle ESS is $1/\sum w_j^2$.

Systematic resampling occurs when ESS falls below .58P or at a forced six-sample move interval. The latter guarantees that both proposal operators are exercised across all conditions. After resampling, one MALA rejuvenation step targets the exact conditional density given the selected ancestor.

This is a resample-move particle filter. Particle ESS measures weight concentration; it is not the Markov-chain ESS formula used for long stationary chains. Because the rejuvenation kernel consists of one step per event, rank-normalized $\hat{R}$ is not meaningful here. We report MALA acceptance and unique ancestors instead.

- Particles $P = 64$.
- Samples $T = 60$.
- Resampling threshold = 37.12 effective particles.
- Forced move interval = 6 samples.
- All posterior summaries are weighted before resampling or equally weighted afterward.

Manifold MALA and Detailed Balance

Let θ contain all Euclidean coordinates and ϕ the phase pair. The proposal mean is $\frac{1}{2}\epsilon^2 s$, with s supplied analytically or by the surrogate. Euclidean coordinates follow $N(\theta + \frac{1}{2}\epsilon^2 s_\theta, \epsilon^2 I)$. Phase coordinates follow the wrapped density induced by Exp_ϕ .

The acceptance probability is $\alpha = \min\{1, \exp[\log \pi(x^*) - \log \pi(x) + \log q(x|x^*) - \log q(x^*|x)]\}$. Both q terms use the scores evaluated at their respective endpoints. The target π is always the exact configured transition times convolution likelihood, including when the surrogate formed the proposal.

For any x and x^* , $\pi(x)q(x^*|x)\alpha(x, x^*) = \min\{\pi(x)q(x^*|x), \pi(x^*)q(x|x^*)\}$; symmetry establishes detailed balance. Retraction wrapping does not invalidate this equality because q is evaluated as a density with respect to product Lebesgue-Haar volume.

Analytical score clipping at ± 120 limits extreme drift. Because the actual clipped score is used in both forward and reverse proposal densities, clipping changes proposal efficiency but not the corrected invariant conditional target.

Analytical Joint Score

Transition scores are closed form. The log-amplitude and frequency terms are Gaussian residual scores. The phase score is the lift-weighted derivative of the wrapped transition. The channel score combines its complex AR prior with the exact observation derivative.

Candidate frequency affects the target twice: through its switching Gaussian transition and through the phase-transition location $\varphi_{t-1} + 2\pi f_t \Delta t$. The implemented frequency score includes $-2\pi\Delta t$ times the phase residual score. Omitting this term would recreate the state-phase contradiction in the original prototype.

For the observation model, differentiation is expressed through real and imaginary directional derivatives. If μ depends on a real coordinate u , then $\partial \log p(y|\mu)/\partial u = -s_R \text{Re}(\partial\mu/\partial u) - s_I \text{Im}(\partial\mu/\partial u)$, where s is the residual score.

A permanent all-coordinate regression test compares the packed analytical score with central finite differences for MP data. In the executed audit, the maximum absolute discrepancy was 7.4×10^{-9} .

Amortized Joint Surrogate

The alternative proposal uses a SiLU random-feature network. Inputs encode standardized innovations for amplitude, frequency, circular phase, padded complex channel residuals, normalized complex observation residual, candidate state, channel-mode indicators, and noise variance.

The hidden map has 72 fixed random SiLU features. Output coefficients are estimated by ridge regression against independently generated exact joint scores. This construction is an amortized differentiable score proxy; it is not treated as a replacement likelihood.

Training used 2214 rows and validation used 486 rows generated independently of the 900 evaluation blocks. Validation score correlation was 0.924, with aggregate score RMSE 12.838.

Metropolis correction protects target invariance from surrogate bias provided the forward and reverse proposal densities use the actual proxy scores. Approximation quality still affects acceptance, mixing, and latency; these are therefore explicit experimental responses.

Physical Simulation Engine

The simulator creates complex I/Q trajectories at 128 Hz. Two emitters have overlapping hop alphabets and stochastic amplitude, frequency, phase, and channel states. The observation channel is LOS, minor multipath, or NLOS tapped delay. The true paths are saved only for evaluation.

Noise is calibrated after clean mixing to nominal +15, +5, or -5 dB. Real and imaginary components each receive an independent Gaussian plus truncated-Cauchy sum. Achieved SNR is retained per run because finite samples and heavy tails make nominal and realized power differ.

Stable keyed seeds separate physical simulation from inference randomization. For a fixed seed, SNR, and channel, both architectures receive the identical observation trajectory and noise hyperparameters. This pairing eliminates between-trajectory variation from architecture contrasts.

Component	Setting
Emitters	2 frequency-hopping complex sources
Length	60 complex I/Q samples
Hop opportunity	Every 12 samples
Particles	64
Physical blocks	900
Method-level records	1,800

Factorial Design

The experiment is a balanced $2 \times 3 \times 3$ full factorial. Factor A is inference architecture. Factor B is nominal SNR (+15, +5, -5 dB). Factor C is channel complexity (LOS, MP, NLOS). Each physical cell contains 100 seed blocks observed under both architectures.

All method contrasts are paired by seed, SNR, and channel. The same seed labels also recur across the nine physical conditions, yielding a repeated-measures block over the complete 18-condition profile. Execution order is deterministic, but latency is measured inside each filter and interpreted as implementation-specific software timing.

The primary response is $\text{MSE}_f = T^{-1} \sum_t \|f_t - \hat{f}_t\|^2$ after the first hop interval. Phase uses circular differences. Channel NMSE divides complex squared error by true channel energy.

Factor	Levels	Role
A: score architecture	2	Primary treatment
B: SNR	3	Ordered environment
C: channel	3	Ordered complexity
Seed block	100	Repeated physical randomization

Blocked Contrast Analysis

For inference, each 18-condition seed vector is projected onto a complete orthonormal factorial basis. Architecture uses a normalized two-level contrast. SNR and channel each use normalized linear and quadratic contrasts. Tensor products generate all main effects and interactions.

For an effect with q contrast coordinates, the 100 seed-level score vectors are tested against zero using Hotelling's T^2 and its exact F transformation: $F = (n-q)T^2/[q(n-1)]$, with q and $n-q$ degrees of freedom. This multivariate repeated-measures test avoids a univariate sphericity assumption.

Frequency MSE and channel NMSE are log transformed; phase RMSE and particle ESS remain on their natural scales; latency is log transformed. Holm adjustment is applied across the seven omnibus effects within each response. Raw paired mean differences and normal intervals remain descriptive summaries on the original scale.

The analysis treats the 100 seed profiles as independent blocks. It does not infer to a population of real receivers or emitters; that external generalization requires a field sampling frame.

Numerical and Structural Verification

Nine automated tests protect the core claims. The convolution density integrates to one, and its score matches central finite differences. The wrapped density integrates to one against torus volume, its score matches finite differences, and phase differences are periodic.

The full joint target test perturbs every packed coordinate-log amplitudes, frequencies, phases, real channel taps, and imaginary channel taps-and compares the analytical gradient with the central-difference derivative. The observed maximum absolute difference was 7.4×10^{-9} .

The structural tests enforce Nyquist-valid overlapping channel alphabets and a blind filter API with no truth or true-channel argument. An NLOS end-to-end smoke test verifies finite outputs and constrains phases to $[0,2\pi)$.

Domain	Tests	Status
Convolution law	Normalization; score derivative	Pass
Torus geometry	Normalization; derivative; periodicity	Pass
Joint target	Nyquist; gradient; API; NLOS smoke	Pass

Primary Frequency-Recovery Results

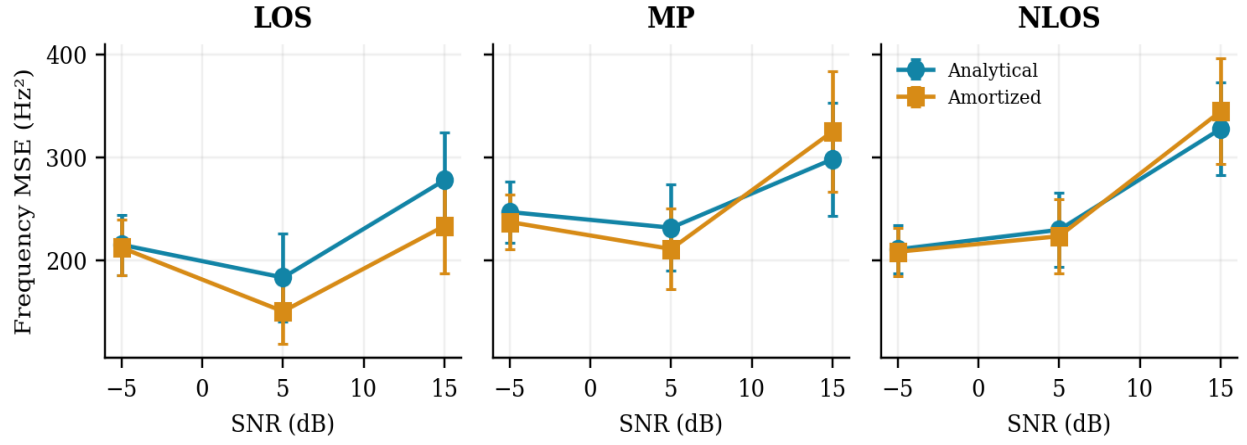


Figure 2. Mean frequency MSE by SNR, channel, and architecture; bars are 95% normal intervals.

Across all conditions, analytical mean MSE was 246.76 Hz² (SD = 207.88); surrogate mean MSE was 238.41 Hz² (SD = 208.73). The paired raw difference was -8.35, 95% CI [-24.83, 8.14].

Thus, the original-scale paired summary does not establish an overall accuracy difference. Error increases as SNR decreases and as the channel moves from LOS to delayed NLOS. Architecture-specific curves largely overlap relative to these environmental shifts.

Phase and Channel Recovery

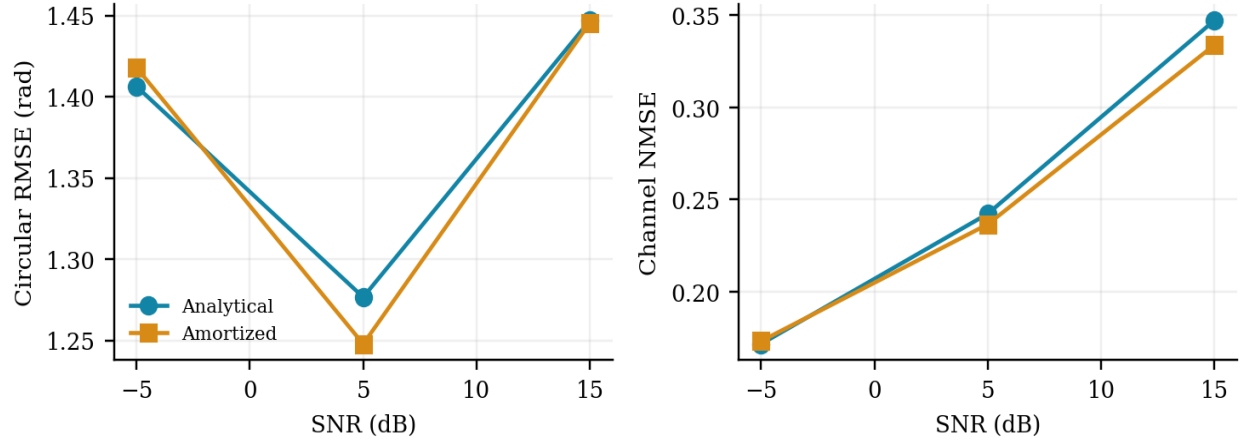


Figure 3. Circular phase RMSE and complex channel NMSE by SNR and architecture.

Mean phase RMSE was 1.377 rad for the analytical score and 1.370 rad for the surrogate. The paired difference was -0.006, 95% CI [-0.035, 0.022].

Mean channel NMSE was 0.254 analytical and 0.248 surrogate. The paired difference was -0.006, 95% CI [-0.015, 0.003]. Neither descriptive interval excludes zero.

Particle Health and Computation

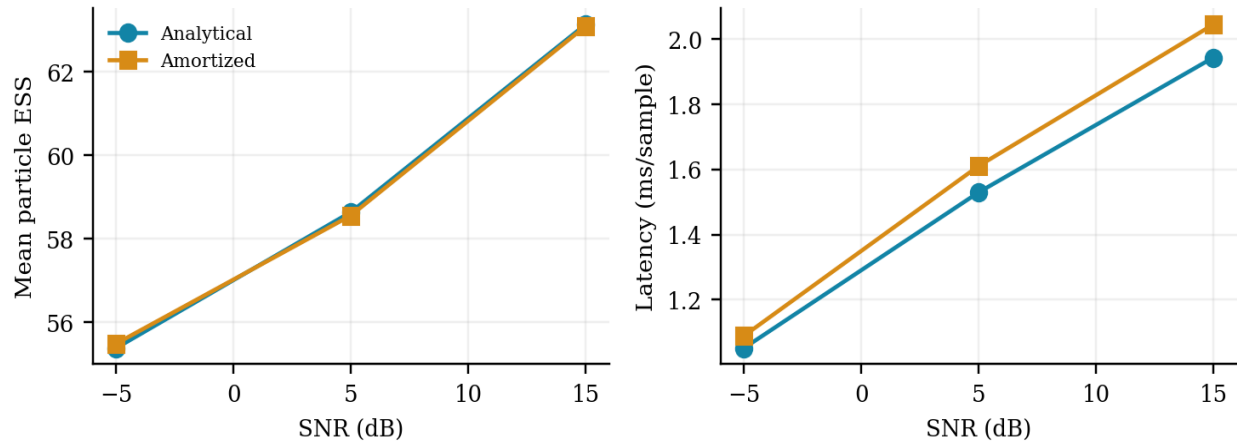


Figure 4. Mean particle ESS and software latency by SNR and architecture.

Particle ESS averaged 59.05 for the analytical method and 59.04 for the surrogate. Their paired difference was -0.006, 95% CI [-0.090, 0.079].

Latency averaged 1.508 ms/sample analytical and 1.581 ms/sample surrogate. The paired difference was 0.073, 95% CI [0.057, 0.090]. Analytical acceptance was 0.974; surrogate acceptance was 0.698.

Factorial Effects

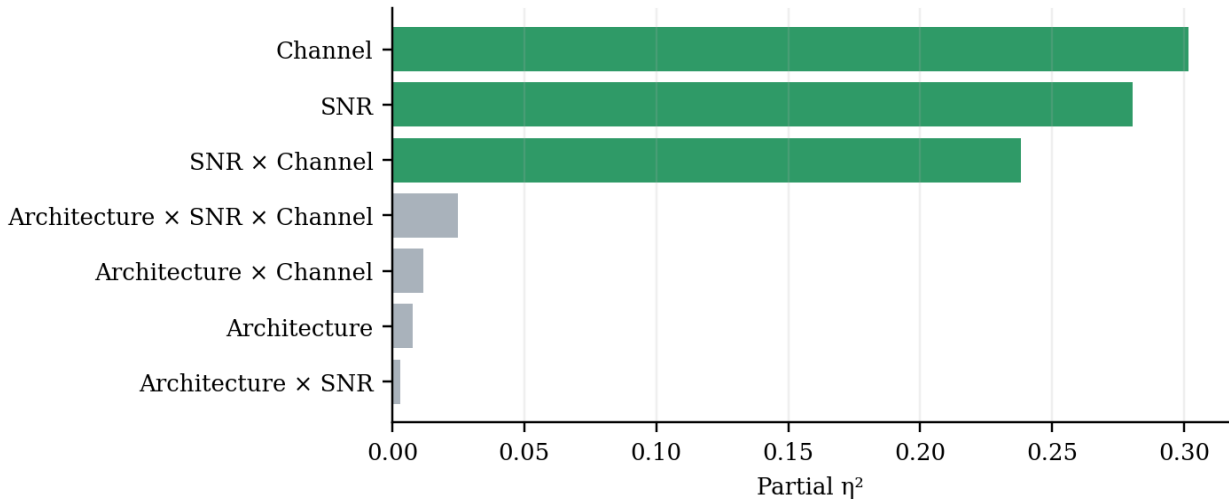


Figure 5. Partial eta squared for omnibus effects on log frequency MSE; green effects survive Holm adjustment.

Effect	F(df)	Holm p	Partial η^2
Architecture	0.78 (1, 99)	1.000	0.008
SNR	19.09 (2, 98)	< .001	0.280
Channel	21.15 (2, 98)	< .001	0.301
Architecture $\times$ SNR	0.14 (2, 98)	1.000	0.003
Architecture $\times$ Channel	0.58 (2, 98)	1.000	0.012
SNR $\times$ Channel	7.50 (4, 96)	< .001	0.238
Architecture $\times$ SNR $\times$ Channel	0.61 (4, 96)	1.000	0.025

Architecture alone yielded $F(1, 99) = 0.78$, Holm $p = 1.000$. SNR, channel, and SNR $\times$ channel survived multiplicity correction. The three-way interaction did not.

Trace-Level Diagnostics and Limitations

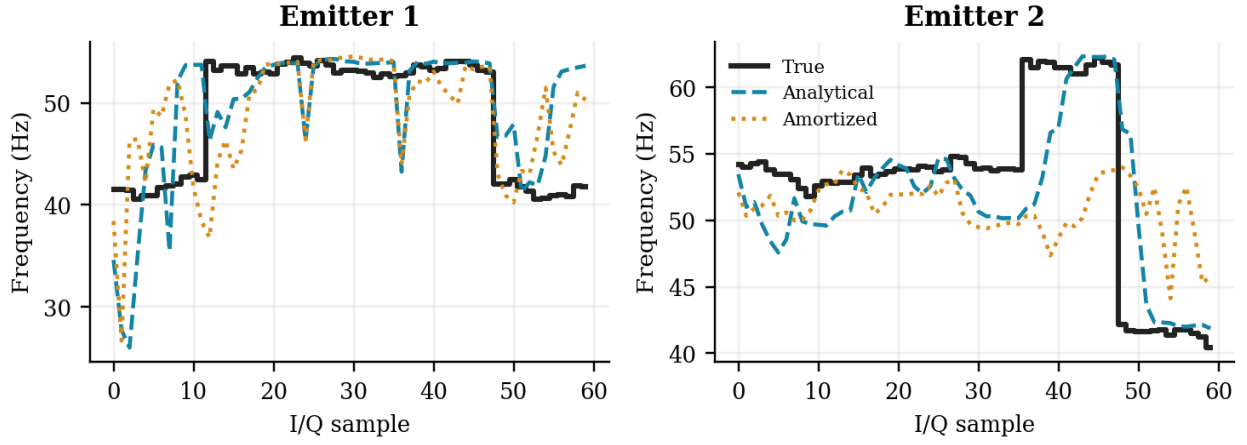


Figure 6. Predeclared seed-0, +5 dB NLOS frequency trace.

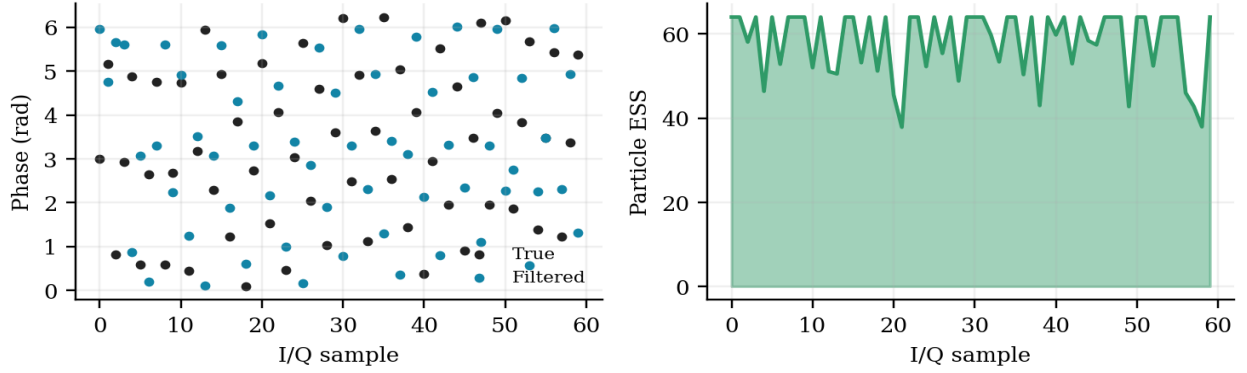


Figure 7. Analytical torus phase and particle ESS for the same trace.

The trace demonstrates execution, not population performance. Posterior means can average competing hop modes and therefore appear between channel centers. Sixty samples and 64 particles are a reproducible benchmark budget, not a sufficiency proof. Label information is partially stabilized by emitter-specific alphabets and dynamics, so unrestricted blind identifiability is not established.

Operational Validation Ladder

Operational validation is an empirical status, not a phrase created by simulator realism. The present study completes numerical verification and representative simulation. It does not include receiver hardware, recorded complex I/Q, lawful controlled emitter schedules, independently maintained truth, or analysts blinded to scenario labels.

Gate	Evidence required	Current status
1. Numerical	Normalization, gradients, invariants	Complete
2. Simulation	Full factorial with latent channel and heavy tails	Complete
3. Hardware-in-loop	Calibrated receiver, clock, quantization, real-time load	Not performed
4. Controlled RF	Recorded I/Q and independent emitter/channel truth	Not performed
5. External replication	Locked code evaluated by independent team	Not performed

A defensible next campaign would preregister scenarios and endpoints, freeze the code and surrogate weights, acquire lawful controlled recordings, separate truth custodians from analysts, measure wall-clock latency on specified hardware, and publish failures as well as successes.

Until those gates are complete, all public language should remain: ‘operationally representative simulation validation.’

Reproducibility and Audit Map

The repository separates generator, inference, analysis, site, and report code. Row-level factorial results are immutable CSV artifacts. Metadata records every configured hyperparameter and the independent surrogate validation summary.

Artifact	Purpose
src/signals_project/joint_ssm.py	Simulator, exact likelihood, surrogate, torus MALA, joint particle filter
tests/test_joint_ssm.py	Eight numerical, geometric, blindness, and smoke tests
data/joint_factorial_results.csv	All 1,800 method-level outcomes
data/joint_cell_summary.csv	Cell means and standard deviations
data/joint_factorial_effects.csv	Blocked multivariate repeated-measures tests
data/joint_example_trace.csv	Predeclared interactive trace
scripts/build_site.py	Self-contained dashboard builder
scripts/build_report.py	This 27-page report builder

Canonical run: `PYTHONPATH=. python -m src.signals_project.joint_ssm --output data --repetitions 100 --particles 64 --samples 60`. Then execute `scripts/analyze_joint_results.py`, `scripts/build_site.py`, and `scripts/build_report.py`.

Conclusions

The reconstruction accomplished the central engineering objective: the broader joint state-space model is no longer merely described. It is executed as a blind resample-move particle filter over switching frequency, torus phase, amplitude, latent complex tapped-delay channels, and source history.

The likelihood now matches the additive noise generator through Gaussian-truncated-Cauchy convolution quadrature. The torus proposal uses tangent increments, exponential retraction, and wrapped forward/reverse densities. Both proposal architectures are corrected against the same exact configured target.

The factorial evidence attributes most log-frequency error variation to SNR, channel complexity, and their interaction. After Holm adjustment, architecture does not show a primary main effect. The amortized score is slower here and accepts fewer rejuvenation moves, despite high held-out score correlation.

These results justify an executed-model and representative-simulation claim. They do not justify operational SIGINT validation. The scientific next step is not stronger wording; it is a preregistered, independently blinded controlled RF campaign on specified hardware.

- Use the analytical score as the current reference implementation.
- Treat the surrogate as an efficiency research branch, not a validated replacement.
- Increase particles and trajectory length in a registered sensitivity study.
- Advance validation only when real I/Q and independent truth are available.

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Repository-specific note

All numerical values in this report are computed from the committed `joint_factorial_results.csv` and associated derived artifacts. References support general methods; none is presented as evidence of operational validation for this implementation.